

Assignment05

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Set up

Packages

```
# general use and cleaning
library(tidyverse)
library(here)
library(janitor)
library(snakecase)

# wrapping axis labels
library(scales)

# reading in .xlsx files
library(readxl)

# visualizing missing data
library(naniar)

# arranging plots
library(patchwork)
```

Data

```
# taxonomic information
taxon_list <- read_csv(here("data", "taxon_list.csv"))
```

```

# aquatic invertebrates
aq_ins <- read_xlsx(here("data", "Aquatic_Sampling_Data-2026-03-10.xlsx"),
  sheet = "Aquatic Insects")

# site information
sites <- read_xlsx(here("data", "Aquatic_Sampling_Data-2026-03-10.xlsx"),
  sheet = "sites")

# water quality information
water_quality <- read_xlsx(here("data",
  ↪ "Aquatic_Sampling_Data-2026-03-10.xlsx"),
  sheet = "Water Quality",
  na = c("", "n/a", "N/A", "over", "173+"))

```

Class code

```

# creating new object from sites
ncos_sites <- sites |>
  # filter to only include NCOS sites sampled four times a year
  filter(sector == "NCOS" & sampling_frequency == "Quarterly")

```

```

# creating new clean object from taxon_list
taxon_list_clean <- taxon_list |>
  # converting taxon name to snake case (no spaces or capital letters,
  # only underscores)
  mutate(verbatim_name = to_any_case(verbatim_name, case = "snake"))

```

```

# creating new clean object from water quality
water_quality_clean <- water_quality |>
  # clean column names
  clean_names() |>
  # filtering join: only include sites in ncos_sites
  semi_join(ncos_sites, by = "site") |>
  # create new column to join with later data frames
  unite("date_site", date, site, remove = TRUE) |>
  # group by date_site column
  group_by(date_site) |>
  # calculate median pH, salinity, DO
  summarize(med_ph = median(p_h, na.rm = TRUE),

```

```

        med_sal = median(salinity_ppt, na.rm = TRUE),
        med_do = median(dissolved_oxygen_mg_l, na.rm = TRUE)) |>
# ungroup data frame
ungroup() |>
# filter out salinity outliers
filter(date_site != "2022-08-12_NVBR")

# creating new clean object from aquatic inverts
aq_ins_clean <- aq_ins |>
# clean column names
clean_names() |>
# filtering join: only include sites occurring in ncos_sites
semi_join(ncos_sites, by = c("site")) |>
# renaming columns
rename(date = date_on_vial) |>
# select columns of interest
select(date, sample_type, site,
        ostracod,
        copepod,
        hemiptera_corixidae_boatman,
        cladocera,
        nematode,
        diptera_ceratopogonidae,
        annelida_oligochaete,
        diptera_chironomid,
        annelida_polychaete) |>
# filter to only include "filtered beaker" samples
filter(sample_type %in% c("FB 250", "FB250")) |>
# convert the data frame to long format
pivot_longer(cols = ostracod:annelida_polychaete,
              names_to = "taxon",
              values_to = "abundance") |>
# group by date, site, taxon
group_by(date, site, taxon) |>
# calculate average abundance per liter (sum abundance divided by 7.5
  ↳ liters)
summarize(ave_lit = sum(abundance, na.rm = TRUE)/7.5) |>
# ungroup data frame
ungroup() |>
# create a new column called `date_site` to join with water quality
unite("date_site", date, site, remove = FALSE) |>
# join with water quality

```

```

left_join(water_quality_clean, by = c("date_site")) |>
# join with taxon list
left_join(taxon_list_clean, by = c("taxon" = "verbatim_name")) |>
# add full names of sites
mutate(site_full = case_when(
  site == "NVBR" ~ "North of Venoco Bridge",
  site == "NEC" ~ "South of Venoco Bridge",
  site == "NMC" ~ "Main Channel",
  site == "NPB" ~ "South of Phelps Bridge",
  site == "NPB1" ~ "North of Phelps Bridge",
  site == "NPB2" ~ "Phelps Road",
  site == "NWP" ~ "West Pond",
  site == "NDC" ~ "Devereux Creek"
)) |>
# setting factor levels for site
mutate(site_full = fct_reorder(site_full, med_sal, .fun = "median", .na_rm
  ↪   = TRUE),
  site_full = fct_rev(site_full))

```

```

# base layer
salinity <- ggplot(data = aq_ins_clean,
  # x-axis: site
  # y-axis: mean salinity
  mapping = aes(x = site_full,
    y = med_sal)) +
# first layer: salinity measurements at each survey
geom_jitter(height = 0,
  width = 0.1,
  shape = 21) +
# wrapping x-axis text
scale_x_discrete(labels = label_wrap(width = 15)) +
# second layer: points to represent means
stat_summary(fun = median,
  color = "red",
  size = 1) +
# relabelling x- and y-axes
labs(x = "Site",
  y = "Median salinity (ppt)",
  title = "Site salinity") +
# different theme
theme_bw()

```

```

# creating new object from clean aquatic inverts
copepods <- aq_ins_clean |>
  # filter to only include copepods
  filter(taxon == "copepod") |>
  # log + 1 transform mean abundance per liter
  mutate(log_ave_lit = log(ave_lit + 1))

# base layer
copepods <- ggplot(data = copepods,
  # x-axis: site
  # y-axis: mean abundance/L
  mapping = aes(x = site_full,
    y = log_ave_lit)) +
  # first layer: points to represent observations
  geom_jitter(height = 0,
    shape = 21,
    width = 0.1) +
  # second layer: points to represent medians
  stat_summary(geom = "point",
    fun = "median",
    size = 4,
    color = "blue") +
  # wrapping x-axis text
  scale_x_discrete(labels = label_wrap(15)) +
  # relabelling x- and y-axis, adding title
  labs(x = "Site",
    y = "log + 1 transformed average abundance/L",
    title = "Copepods") +
  # different theme
  theme_bw()

```

1. Visualizing differences across sites in major taxon abundance

a. Plan your figure

- x-axis: different sites
- y-axis: log + 1 transformed average ostracod abundance per liter
- geometry to represent average abundance/L: `geom_jitter()`, circle points
- geometry to represent medians: `stat_summary()`, circle points with a distinct color

b. Wrangle your data

```
# create new object from cleaned aquatic invertebrate data
ostracods_df <- aq_ins_clean |>
  # keep only ostracod rows
  filter(taxon == "ostracod") |>
  # log + 1 transform average abundance per liter
  mutate(log_ave_lit = log(ave_lit + 1))

# display 10 random rows
ostracods_df |>
  slice_sample(n = 10)
```

```
# A tibble: 10 x 24
```

	date_site <chr>	date <dtm>	site <chr>	taxon <chr>	ave_lit <dbl>	med_ph <dbl>	med_sal <dbl>	med_do <dbl>
1	2023-05-06_NDC	2023-05-06 00:00:00	NDC	ostr~	56	10.1	NA	13.1
2	2022-02-23_NPB1	2022-02-23 00:00:00	NPB1	ostr~	0	NA	NA	NA
3	2023-03-03_NDC	2023-03-03 00:00:00	NDC	ostr~	5.73	7.45	0.688	5.64
4	2023-02-21_NEC	2023-02-21 00:00:00	NEC	ostr~	0	8.56	NA	8.8
5	2023-09-12_NVBR	2023-09-12 00:00:00	NVBR	ostr~	0	9.14	NA	20.3
6	2020-02-21_NVBR	2020-02-21 00:00:00	NVBR	ostr~	4.8	7.34	4.1	6.1
7	2022-11-12_NEC	2022-11-12 00:00:00	NEC	ostr~	0	7.33	7.01	2.4
8	2023-02-12_NEC	2023-02-12 00:00:00	NEC	ostr~	2.13	NA	NA	NA
9	2022-08-15_NMC	2022-08-15 00:00:00	NMC	ostr~	0.133	7.61	NA	NA
10	2023-09-15_NMC	2023-09-15 00:00:00	NMC	ostr~	0.4	NA	NA	14.2

```
# i 16 more variables: taxonRank <chr>, scientificName <chr>, kingdom <chr>,
#   Phylum <chr>, Class <chr>, Subclass <chr>, Superorder <chr>, Order <chr>,
#   Infraorder <chr>, Family <chr>, Subfamily <chr>, Tribe <chr>, Genus <chr>,
#   comments <chr>, site_full <fct>, log_ave_lit <dbl>
```

c. Code your figure

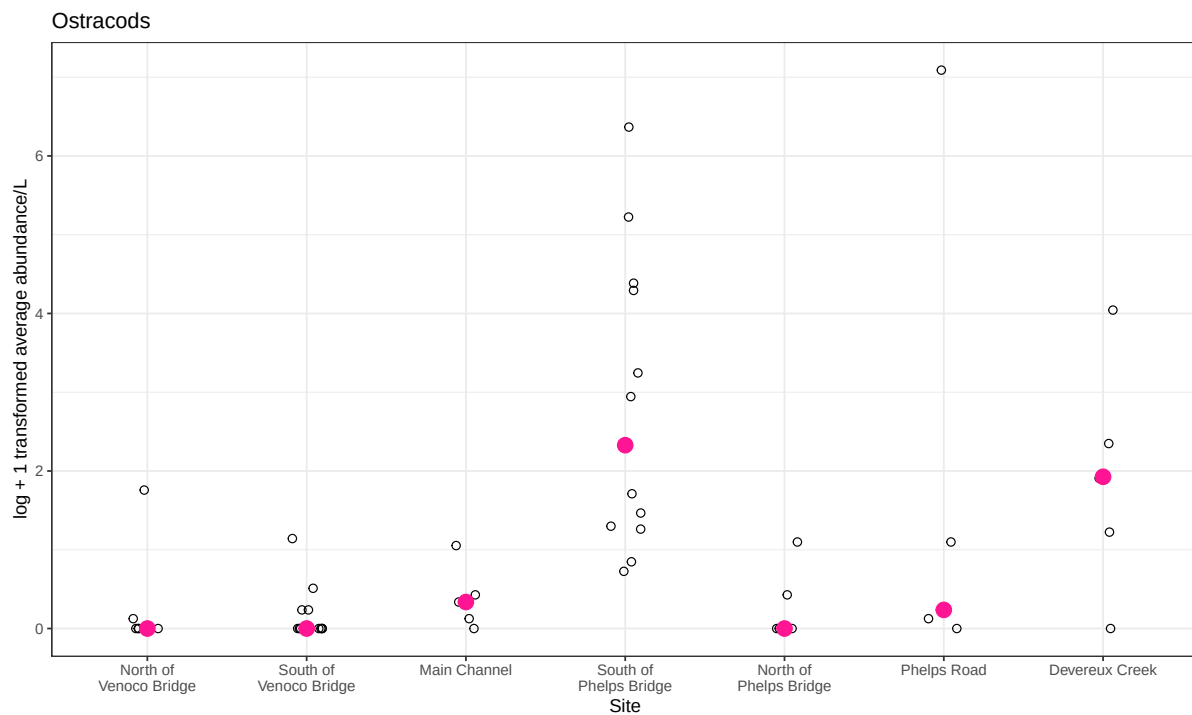
```
# base layer
ostracods <- ggplot(data = ostracods_df,
  # x = site, y = log abundance/L
  mapping = aes(x = site_full,
    y = log_ave_lit)) +
  # first layer: individual survey points
```

```

geom_jitter(height = 0,
            width = 0.1,
            shape = 21,
            size = 2) +
# second layer: median per site, distinct in size and color
stat_summary(fun = "median",
            geom = "point",
            color = "deeppink",
            size = 4) +
# wrap long site labels for readability
scale_x_discrete(labels = label_wrap(15)) +
# readable axis labels and taxon-specific title
labs(x = "Site",
     y = "log + 1 transformed average abundance/L",
     title = "Ostracods") +
# non-default theme
theme_bw()

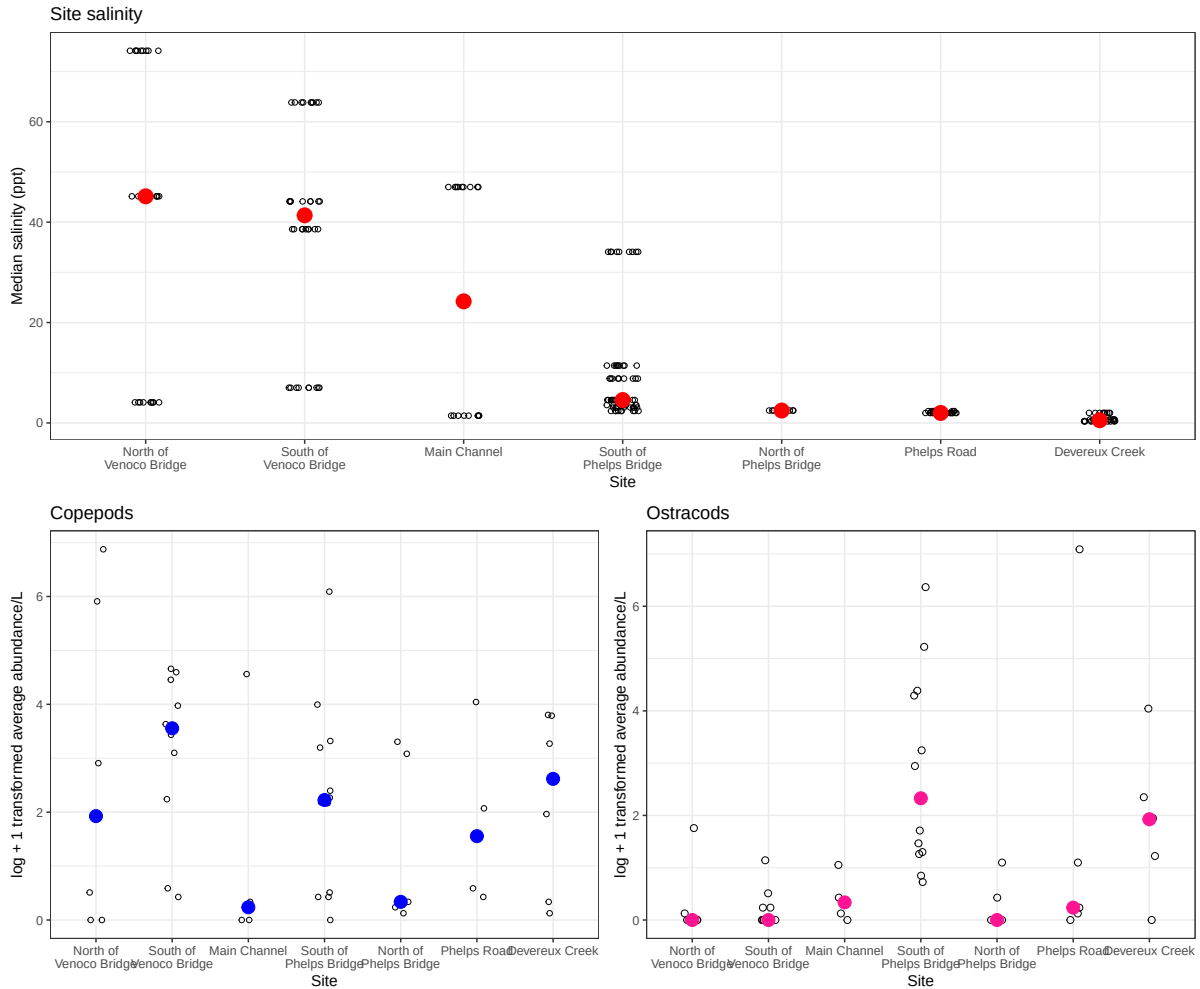
# display object
ostracods

```



d. Create a multipanel figure

```
# salinity on top, copepods and ostracods side-by-side below  
(salinity) / (copepods | ostracods)
```



e. Interpret your figure

- Differences between sites in copepod and ostracod abundance: Copepod medians are the blue circles on the bottom left panel. They are generally non-zero at every site. They have largest at South of Venoco Bridge (about 3.5) and the lowest at Main Channel and North of Phelps Bridge (~0.3). Ostracod medians are the pink circles on the bottom right panel. They are essentially zero at the two Venoco Bridge sites and at North of Phelps Bridge. They have sharply risen at South of Phelps Bridge and Devereux Creek.

- How ostracod abundance may relate (or not) to site-level differences in salinity: Because sites are ordered from highest to lowest median salinity left-to-right in the top panel, the pink ostracod medians generally rise as the red salinity medians fall. Ostracods are zero where median salinity exceeds 40 ppt and reach their highest values at sites with median salinity below 5 ppt. The relationship isn't very correlated. Main Channel, North of Phelps Bridge, and Phelps Road all have low ostracod medians despite moderate or low salinity, so salinity alone doesn't fully explain the site-level pattern.

Problem 2. Visualizing total abundance as a function of salinity

a. Plan your figure

- x-axis: median salinity, `med_sal`, ppt
- y-axis: log + 1 transformed average abundance per liter, `log_ave_lit`
- color: `taxon`, each taxon its own color
- geometry: `geom_point()`, circle points
- facets: `facet_wrap(~ taxon, scales = "free")`

b. Wrangle your data

```
# create new object from cleaned aquatic invertebrate data
abundance_salinity <- aq_ins_clean |>
  # keep only the columns needed for the figure
  select(date_site, site_full, taxon, ave_lit, med_sal) |>
  # log + 1 transform the abundance per liter
  mutate(log_ave_lit = log(ave_lit + 1)) |>
  # convert taxon names to title case
  mutate(taxon = str_replace_all(taxon, "_", " "),
         taxon = str_to_title(taxon)) |>
  # order taxon factor levels by maximum abundance
  mutate(taxon = fct_reorder(taxon, ave_lit,
                             .fun = max, .na_rm = TRUE,
                             .desc = TRUE))

# show 10 random rows
abundance_salinity |>
  slice_sample(n = 10)
```

```
# A tibble: 10 x 6
```

	date_site	site_full	taxon	ave_lit	med_sal	log_ave_lit
	<chr>	<fct>	<fct>	<dbl>	<dbl>	<dbl>
1	2023-05-04_NEC	South of Venoco Bridge	Nematode	0	NA	0
2	2022-08-15_NMC	Main Channel	Diptera C~	0	NA	0
3	2023-05-05_NPB	South of Phelps Bridge	Hemiptera~	0	NA	0
4	2024-01-28_NPB	South of Phelps Bridge	Annelida ~	0	3.08	0
5	2024-01-28_NPB	South of Phelps Bridge	Diptera C~	0	3.08	0
6	2023-01-28_NPB1	North of Phelps Bridge	Diptera C~	0	NA	0
7	2022-02-25_NEC	South of Venoco Bridge	Nematode	0	44.1	0
8	2023-01-28_NPB1	North of Phelps Bridge	Hemiptera~	0	NA	0
9	2022-11-18_NVBR	North of Venoco Bridge	Copepod	0.667	NA	0.511
10	2022-11-19_NDC	Devereux Creek	Cladocera	0	NA	0